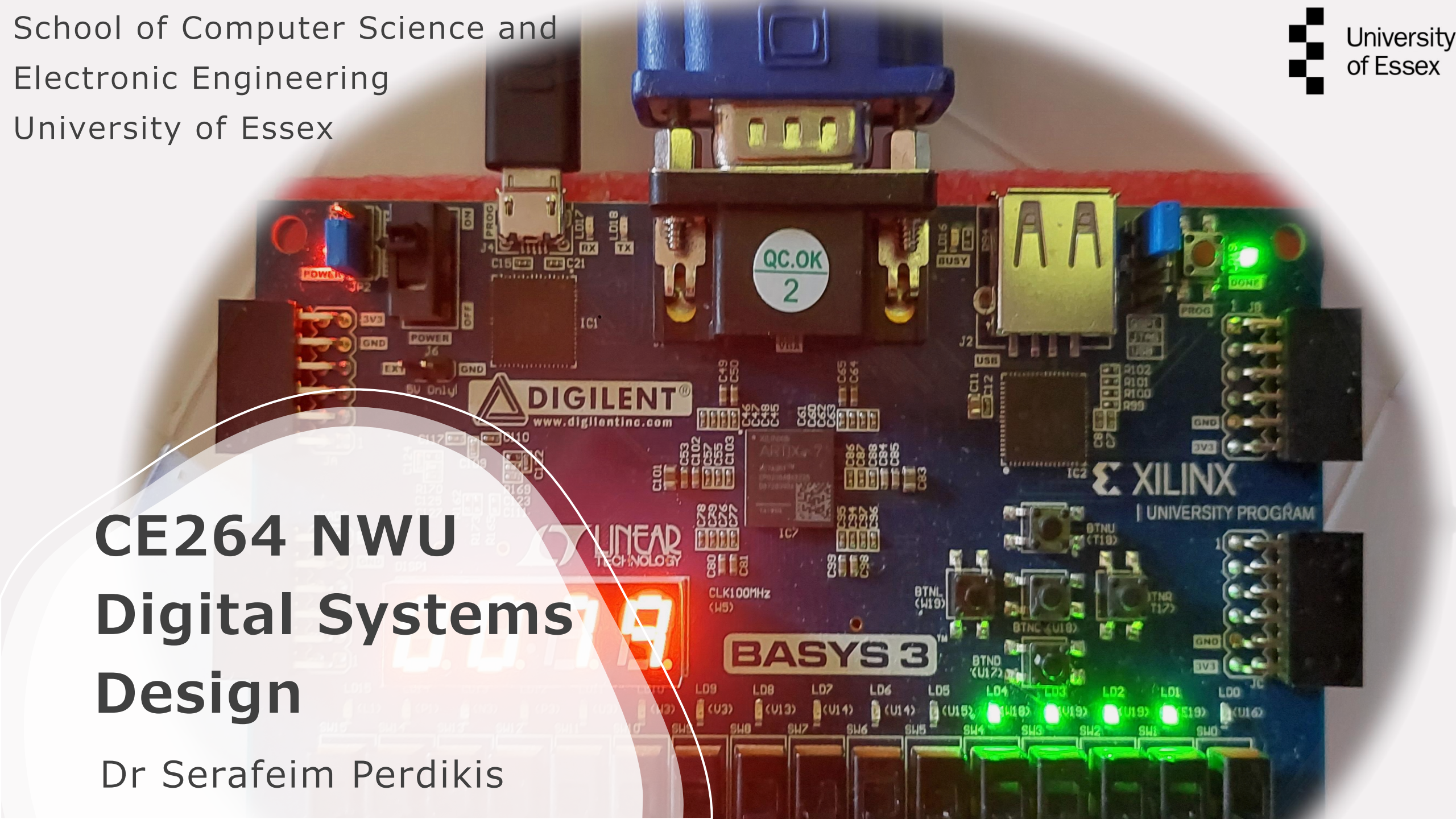
School of Computer Science and
Electronic Engineering
University of Essex

University of Essex

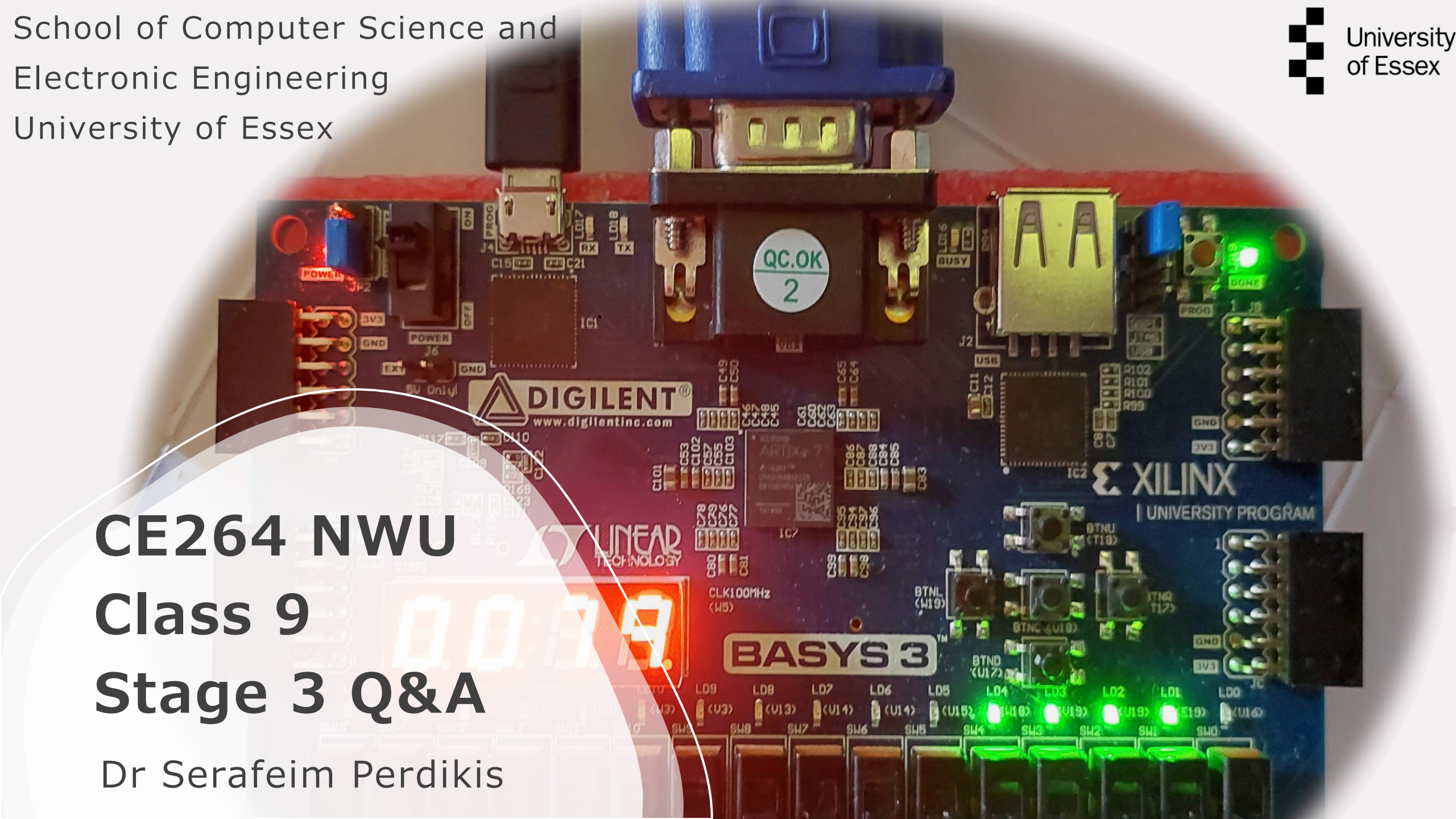


CE264 NWU Digital Systems Design

Dr Serafeim Perdikis

School of Computer Science and
Electronic Engineering
University of Essex

University of Essex

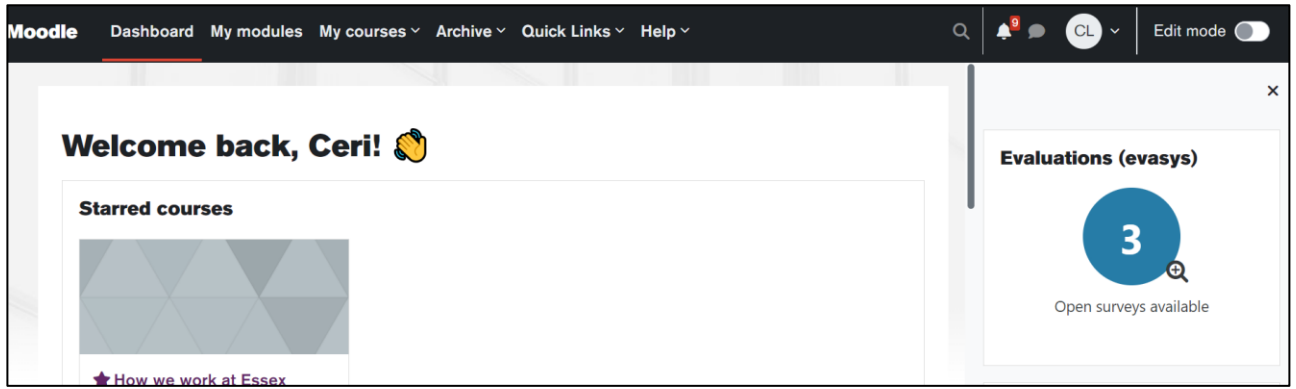


CE264 NWU
Class 9
Stage 3 Q&A

Dr Serafeim Perdikis

Student Module Feedback

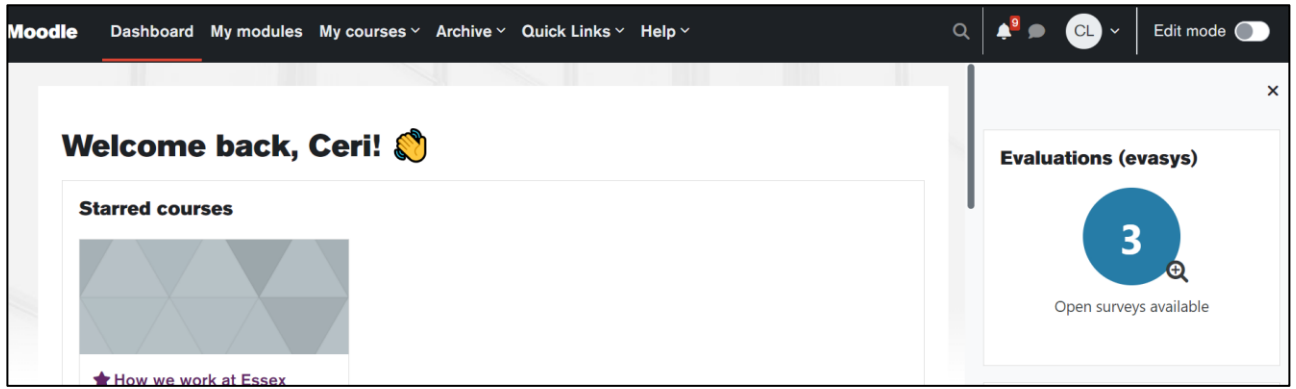
- What's that?
  - Questionnaire for students to evaluate a module
  - Necessary for teachers and TAs to improve delivery!!!
- How do I complete that?
  - You should have received an email to your Essex email address from 'University of Essex surveys' with links to all the surveys available to you.
  - You can also access your available surveys via Moodle:



This will only show if there are surveys available to you.

Student Module Feedback

- Please, take 10' to complete this questionnaire **now**!
  - Log into your Essex email and find the email sent to you (see below)
  - Log into Essex Moodle and look for “evasys” in home screen (see below)
- How do I complete that?
  - You should have received an email to your Essex email address from ‘University of Essex surveys’ with links to all the surveys available to you.
  - You can also access your available surveys via Moodle:



This will only show if there are surveys available to you.

CE264 Social Event

- Tomorrow Tuesday 2nd April at 17:00
  - Held in a classroom in campus (to be announced shortly)
- Attending Y4 in Essex:
  - Providing essential info about the University of Essex and Colchester
  - Discuss advantages
  - Answer your questions
- Please join me! ☺

Demonstrating your progress

- In the next (and last) two labs, show me your progress!
- Call me to your desk when you accomplish any of these:
  - Stage 2: able to show colour on the screen with switches
  - Stage 3, feature 1: able to show a rectangle on a screen
  - Stage 3, feature 2: able to move the rectangle
  - Stage 3, feature 3: able to show coordinates (LEDS, 7-segment display)
  - Stage 3, feature 4: able to store two colours in memory
  - Stage 3, feature 5: able to change/flicker colour of rectangle
- **This will not be marked directly** (not part of marking criteria)
  - …but it will "condition me" positively when marking!

Lab/material access after April 6th

- Access to the labs (Vivado, Multisim, oscilloscope) and material (Basys 3 board, VGA monitor) will be given after Week 4 and until May 7th (submission day)
- Possibility to:
  - Continue your design with Stage 3 and additional features (open-ended)
  - Test your design and shoot the testing videos required
- Access will be limited so:
  - Focus and push yourselves to finish as much as possible within Week 4
  - Plan well so that you can have designs ready for testing on access day

FAQ

- VGA standards: use the Table in slides and Basys 3 manual only
- Testing Stage 2: "…*Example: use the most significant bit of VCOUNT to drive an RGB signal and display two colours, splitting the screen horizontally at some point (exactly where isn't important).*"
  - Not needed! Just a recommendation for good testing for you to be sure your VGA driver is good
  - Marking only based on showing colour on the whole screen with switches
- Testing Stage 2: "Video evidence of RGB **and of VCOUNT / HCOUNT testing**"
  - Ignore the VCOUNT/HCOUNT testing (is done in Stage 3)
- Testing HSYNC/VSYNV/BLANKING with oscilloscope
  - Feel free to test BLANKING as a whole, or HBLANK/VBLANK separately
  - If oscilloscope signal does not look right, report it and attempt explanation
- Submission of unit-test schematics and Constraints.xdc files:
  - For **each testing video** submit both:
    - the Multisim schematic *.ms14 AND
    - The constraints file *.xdc that allowed you to shoot the video
  - Submit any other unit-test schematics & constraints you did, even if it did not led to a video

Revision

- Re-download the slides for revision:
  - Corrections and
  - Improvements have been effectuated
  - Check also relevant emails in your Essex account
- Let’s browse the slides together:
  - Pointing out important/unimportant topics for Progress Test and Exam

Stage 3 design

- Questions?
- Which are the essential components for each feature?

Thank you!

- Questions?